

TITLE:

PHQ/GAD score distribution analysis

Dataset Used:

anxiety_depression_data.csv

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SUBMITTED TO:

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Dataset Introduction

The dataset used for this project is `anxiety_depression_data.csv`, which contains mental health assessment information related to anxiety and depression disorders. The dataset includes demographic features such as:

- Age
- Gender
- Anxiety Scores (GAD)
- Depression Scores (PHQ)

The main objective of this analysis is to study how anxiety and depression scores are distributed across different genders and age groups. Statistical visualizations such as **box plots** and **violin plots** help identify:

- Median scores
- Score spread and variability
- Outliers
- Distribution density
- Age and gender-based mental health trends

This analysis can support mental health research, healthcare planning, and public health insights.

Technologies Used

Python

Python is the main programming language used for data analysis and visualization because of its simplicity and powerful data science libraries.

Python

Uses:

- Data processing
- Statistical analysis
- Visualization creation
- Data grouping and filtering

Pandas

Pandas is a Python library used for handling structured datasets.

pandas

Uses:

- Reading CSV files
- Data cleaning
- Grouping and aggregation
- Creating summary statistics

Matplotlib

Matplotlib is a visualization library used to generate charts and plots.

Matplotlib

Uses:

- Box plots
- Violin plots
- Distribution visualization
- Statistical graph generation

Google Colab Notebook

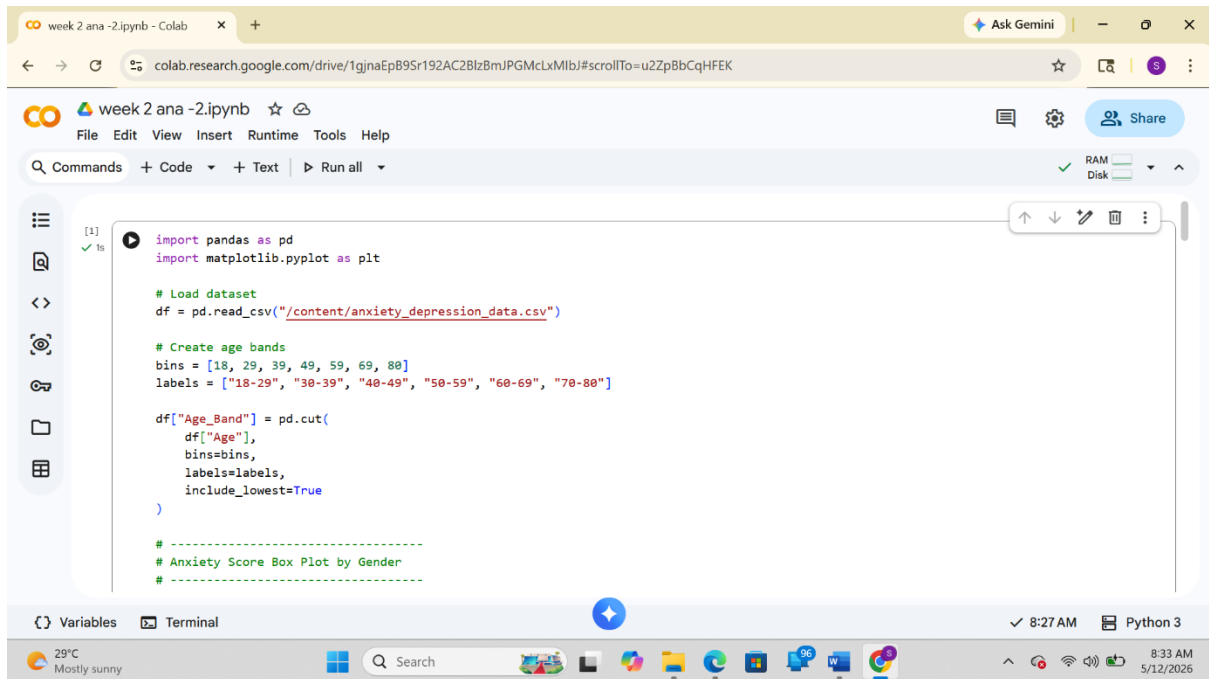
The analysis platform used is Google Colab Notebook.

Google Colab

Advantages:

- Cloud-based execution
- No software installation required
- Easy dataset upload
- Supports Python libraries directly
- Free GPU/CPU access

Score Distribution Analysis



The screenshot shows a Google Colab notebook titled "week 2 ana -2.ipynb". The code in the first cell is as follows:

```
[1] ✓ 1s
import pandas as pd
import matplotlib.pyplot as plt

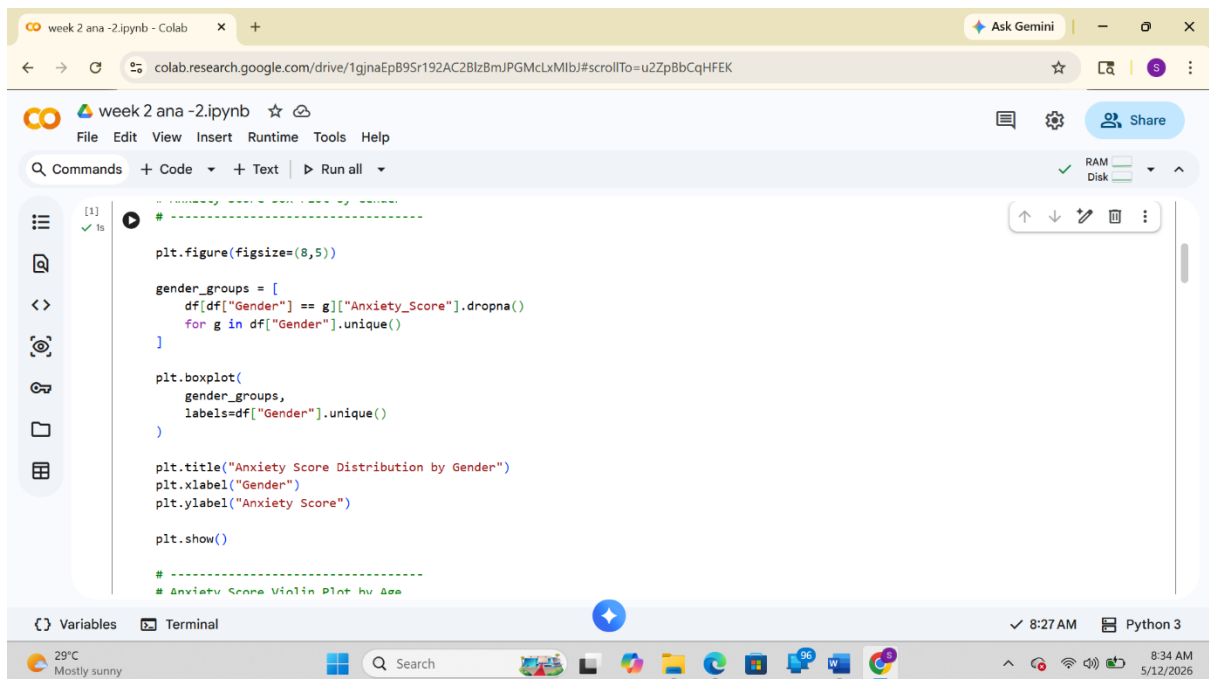
# Load dataset
df = pd.read_csv("/content/anxiety_depression_data.csv")

# Create age bands
bins = [18, 29, 39, 49, 59, 69, 80]
labels = ["18-29", "30-39", "40-49", "50-59", "60-69", "70-80"]

df["Age_Band"] = pd.cut(
    df["Age"],
    bins=bins,
    labels=labels,
    include_lowest=True
)

# -----
# Anxiety Score Box Plot by Gender
# -----
```

The interface includes a left sidebar with navigation icons, a top menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help", and a bottom status bar showing "8:27 AM" and "Python 3".



The screenshot shows the same Google Colab notebook with the second cell of code. The code is as follows:

```
[1] ✓ 1s
# -----
# Anxiety Score Box Plot by Gender
# -----

plt.figure(figsize=(8,5))

gender_groups = [
    df[df["Gender"] == g]["Anxiety_Score"].dropna()
    for g in df["Gender"].unique()
]

plt.boxplot(
    gender_groups,
    labels=df["Gender"].unique()
)

plt.title("Anxiety Score Distribution by Gender")
plt.xlabel("Gender")
plt.ylabel("Anxiety Score")

plt.show()

# -----
# Anxiety Score Violin Plot by Age
# -----
```

The interface is consistent with the previous screenshot, showing the same menu and status bar.

Colab interface showing a Jupyter Notebook titled "week 2 ana -2.ipynb". The code cell [1] contains the following Python code:

```
# -----  
# Anxiety Score Violin Plot by Age  
# -----  
  
plt.figure(figsize=(10,5))  
  
age_groups = [  
    df[df["Age_Band"] == band]["Anxiety_Score"].dropna()  
    for band in labels  
]  
  
plt.violinplot(  
    age_groups,  
    showmeans=True,  
    showmedians=True  
)  
  
plt.xticks(range(1, len(labels)+1), labels)  
  
plt.title("Anxiety Score Distribution by Age Band")  
plt.xlabel("Age Band")  
plt.ylabel("Anxiety Score")
```

The interface includes a left sidebar with navigation icons, a top bar with "Ask Gemini" and "Share" buttons, and a bottom status bar showing "8:27 AM Python 3" and "29°C Mostly sunny".

Colab interface showing a Jupyter Notebook titled "week 2 ana -2.ipynb". The code cell [1] contains the following Python code:

```
# -----  
# Depression Score Box Plot by Gender  
# -----  
  
plt.figure(figsize=(8,5))  
  
gender_groups_dep = [  
    df[df["Gender"] == g]["Depression_Score"].dropna()  
    for g in df["Gender"].unique()  
]  
  
plt.boxplot(  
    gender_groups_dep,  
    labels=df["Gender"].unique()  
)  
  
plt.title("Depression Score Distribution by Gender")  
plt.xlabel("Gender")  
plt.ylabel("Depression Score")  
  
plt.show()
```

The interface includes a left sidebar with navigation icons, a top bar with "Ask Gemini" and "Share" buttons, and a bottom status bar showing "8:27 AM Python 3" and "29°C Mostly sunny".

Colab interface showing a Jupyter Notebook titled "week 2 ana -2.ipynb". The code cell contains the following Python code:

```
[1] ✓ ts
# Depression Score Violin Plot by Age
# -----

plt.figure(figsize=(10,5))

age_groups_dep = [
    df[df["Age_Band"] == band]["Depression_Score"].dropna()
    for band in labels
]

plt.violinplot(
    age_groups_dep,
    showmeans=True,
    showmedians=True
)

plt.xticks(range(1, len(labels)+1), labels)

plt.title("Depression Score Distribution by Age Band")
plt.xlabel("Age Band")
plt.ylabel("Depression Score")
```

The interface includes a left sidebar with navigation icons, a top bar with "Ask Gemini" and "Share" buttons, and a bottom status bar showing "8:27 AM Python 3" and "29°C Mostly sunny 5/12/2026".

Colab interface showing a Jupyter Notebook titled "week 2 ana -2.ipynb". The code cell contains the following Python code:

```
[1] ✓ ts
showmeans=True,
showmedians=True
)

plt.xticks(range(1, len(labels)+1), labels)

plt.title("Depression Score Distribution by Age Band")
plt.xlabel("Age Band")
plt.ylabel("Depression Score")

plt.show()

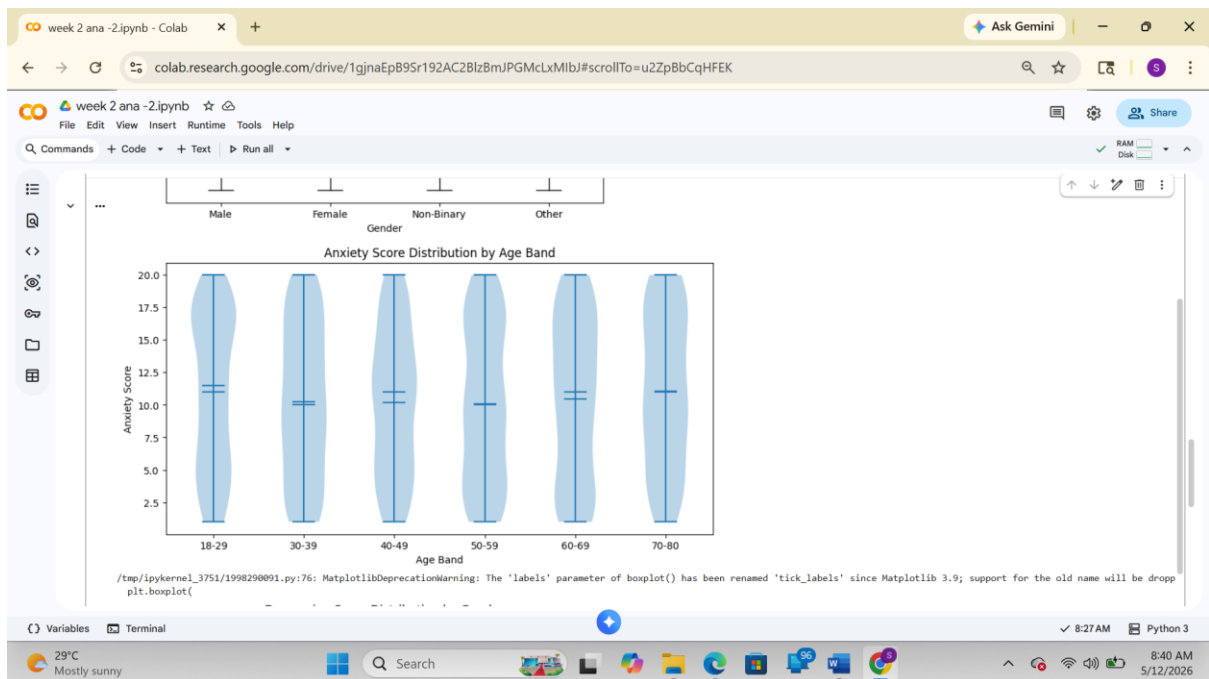
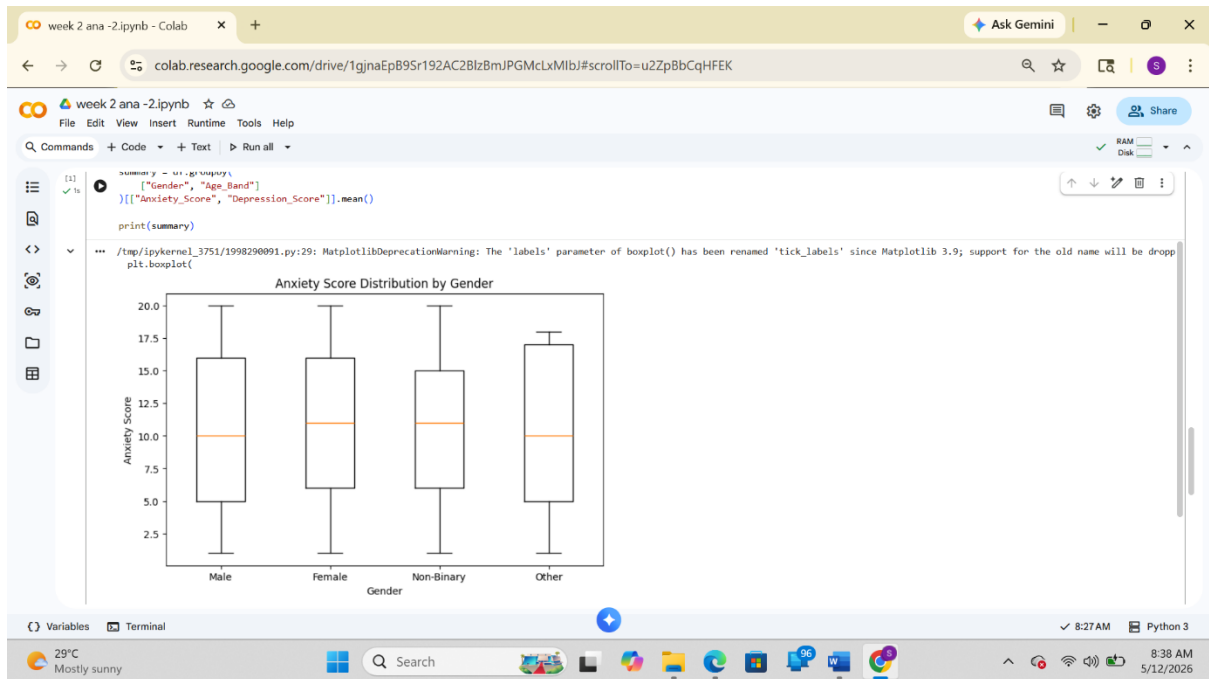
# -----
# Summary Statistics
# -----

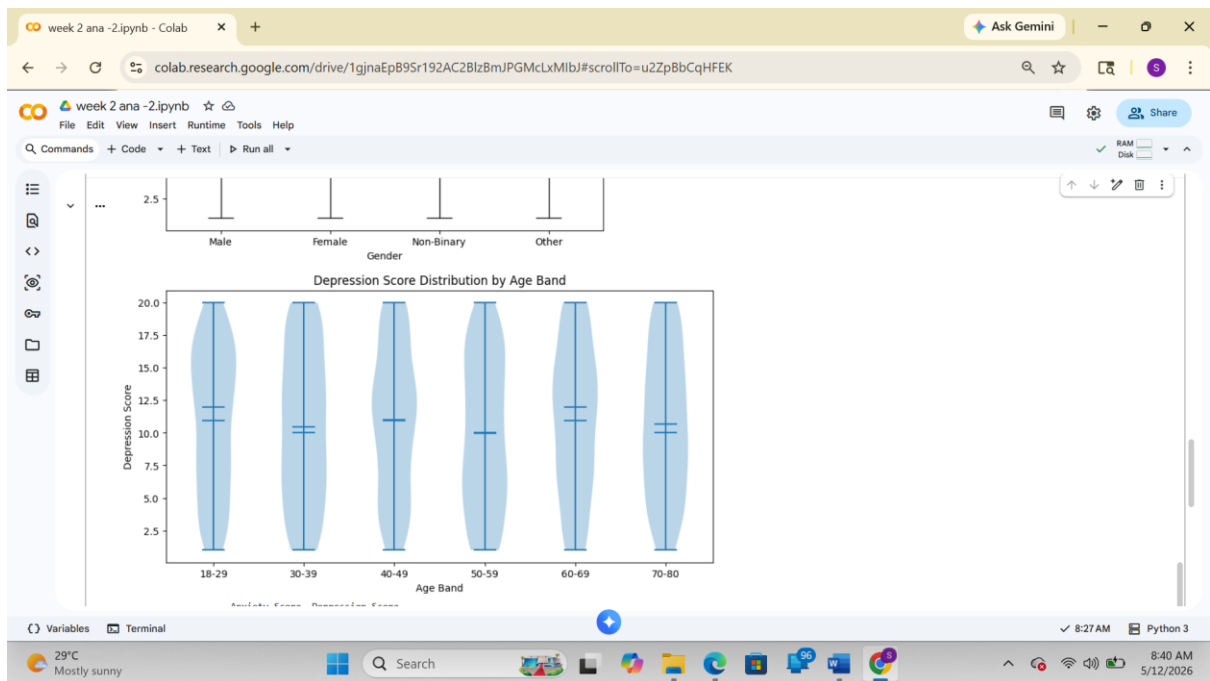
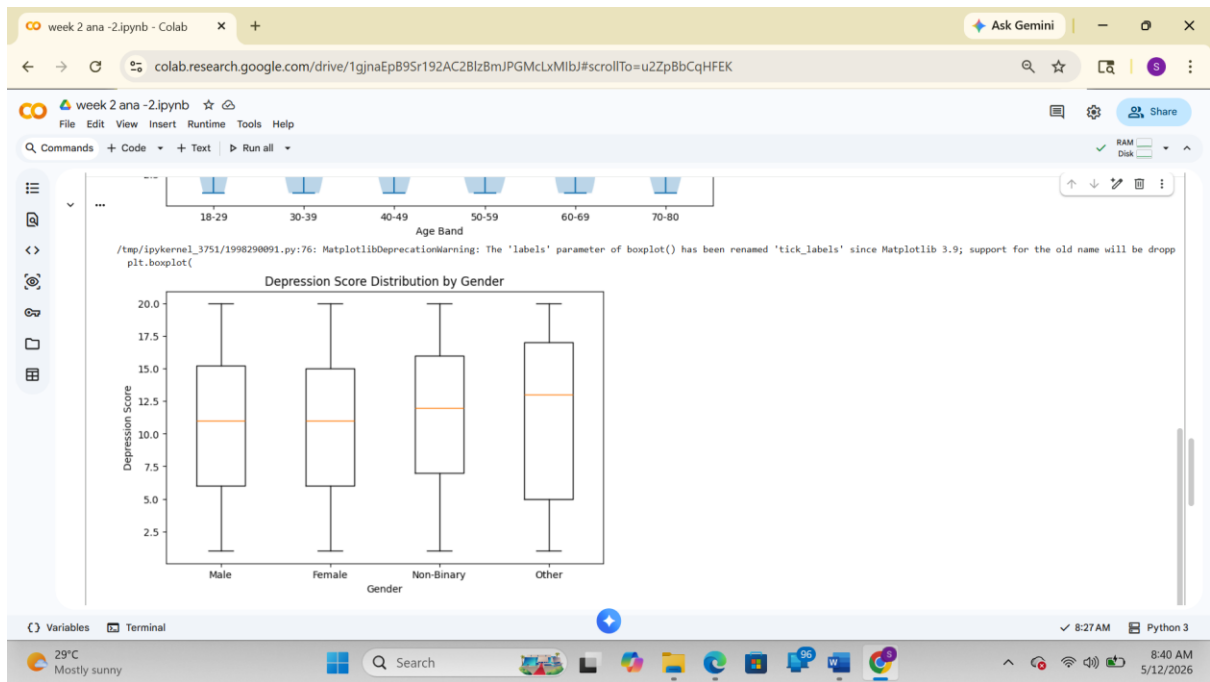
summary = df.groupby(
    ["Gender", "Age_Band"]
)[["Anxiety_Score", "Depression_Score"]].mean()

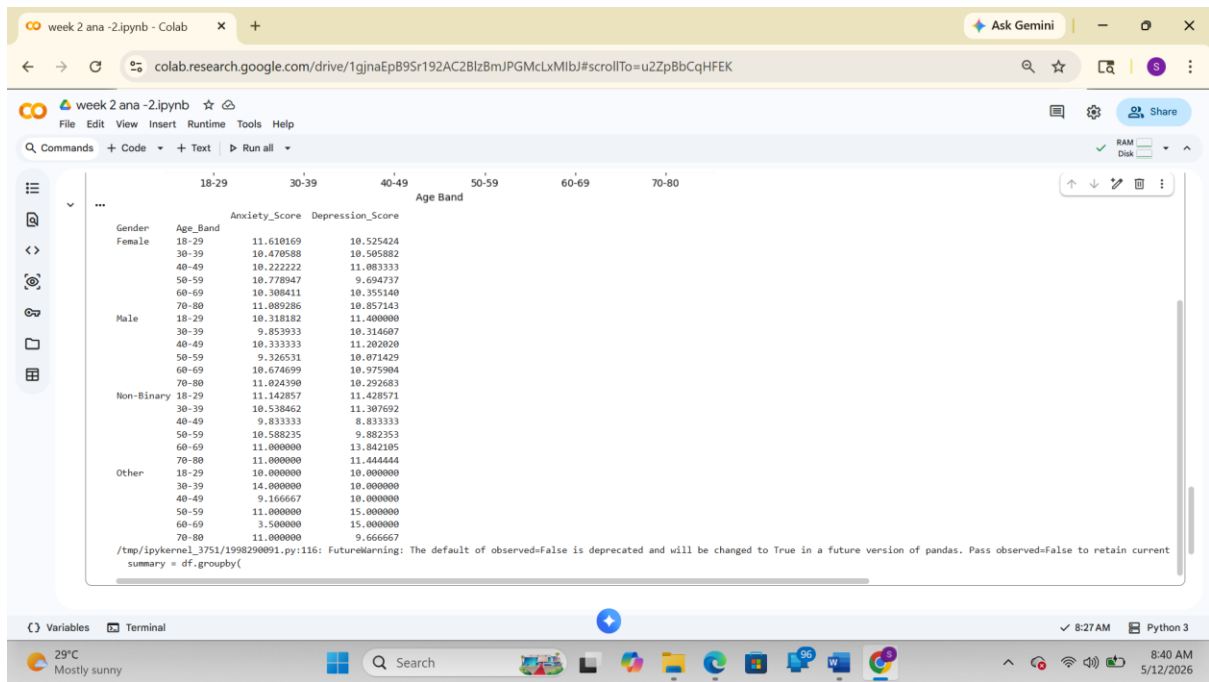
print(summary)
```

The interface includes a left sidebar with navigation icons, a top bar with "Ask Gemini" and "Share" buttons, and a bottom status bar showing "8:27 AM Python 3" and "29°C Mostly sunny 5/12/2026".

Output







After loading the dataset, the data was grouped according to gender and age categories to better understand mental health score patterns. The analysis focused on two important mental health indicators:

- **Anxiety Scores (GAD)**
- **Depression Scores (PHQ)**

Age values were converted into meaningful age bands such as 18–29, 30–39, and so on. This helped compare mental health score distributions across different life stages.

Box Plot Analysis

Box plots were used to visualize:

- Median scores
- Minimum and maximum values
- Interquartile ranges
- Outliers

These plots clearly showed how anxiety and depression scores differed between genders. Outliers indicated individuals with unusually high or low mental health scores.

Violin Plot Analysis

Violin plots provided a deeper understanding of score distributions by showing:

- Data density

- Distribution shape
- Concentration of scores
- Spread of values

The violin plots helped identify which age groups had higher variability in anxiety and depression levels.

Conclusion

The PHQ/GAD distribution analysis provided important insights into mental health score patterns across age and gender groups.

Key Findings:

- Anxiety and depression scores vary significantly across demographic categories.
- Box plots helped identify:
 - Median score levels
 - Interquartile ranges
 - Outliers in mental health scores
- Violin plots clearly showed:
 - Score density
 - Distribution shape
 - Variability among age groups
- Some age groups displayed wider score distributions, indicating higher variability in mental health conditions.
- Gender-based differences in anxiety and depression patterns were also observable.

Overall Outcome:

This analysis demonstrates how data visualization techniques can effectively explore mental health disorder trends and support evidence-based healthcare research using Python and Google Colab.